

Health and Economic Returns to Dietary Sodium Reduction

A national-scale cardiovascular disease modeling and benefit–cost analysis

Resolve to Save Lives (RTSL) – UW

July 2026

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Study Overview

Project Objective and Policy Impact

- Estimate national-scale health impacts of dietary sodium reduction policies using a population CVD model.
- Quantify deaths and cases of ischemic heart disease (IHD), stroke, and hypertensive heart disease, as well as DALYs averted over 2026–2050.
- Conduct a full economic benefit–cost analysis to support government decision-making.
- Focus on RTSL partner countries and on policies that governments can realistically implement.

Agenda for Today's Call

1. **Walk through the 6 interventions** in the Sodium Lives Saved Calculator — confirm effect sizes and source assignments.
2. **Matti (if present)** discuss discretionary-vs-packaged disaggregation approach — GDP regression, floor/ceiling values, BP effects of K and Na.
3. **Decide pending items 1–4:** LSSS scenarios, mass media inclusion, observation-period harmonization, country-specific vs. regression-based sodium shares.
4. **Lock revised timeline** for paper + report + Bloomberg renewal deliverable.

Recap of Prior Presentation

Where We Left Off

- **Modeling framework:** Multi-state CVD life-table (Well → Sick → Dead) simulated annually over 2026–2050 for ages 20–95, four CVD causes (IHD, stroke, HHD, other CVD), by age × sex × country.
- **Outcome metrics:** Deaths averted, IHD/stroke/HHD cases averted, DALYs averted, plus full benefit–cost analysis (NPV, BCR).
- **Baseline scenario design:** Five SHAKE-aligned policy bundles with total sodium-reduction efficacy $\text{salteff}(\eta)$ ranging from 0 (baseline) to 0.250 (full mandatory package).
- **Causal pathway:** Sodium → SBP (Filippini dose–response) → CVD incidence (Ettehad per-10-mmHg RRs) → life-table transitions.
- **Deliverables:** Peer-reviewed paper + RTSL report + interactive Shiny tool, originally targeted for end of June 2026 (per Apr 24 timeline).

Sources of Sodium — Current State

Zina and Nicole have been compiling literature on sodium shares across three categories:

- **Packaged foods:** Processed and ultra-processed sources, including salt-based condiments (fish sauce, soy sauce) where separately reported.
- **Discretionary sources:** Salt or MSG added while cooking at home or at the table.
- **Outside-the-home:** Restaurants, cafeterias/canteens, street food. Approximately $\frac{1}{3}$ of outside-home assumed to be institutional (public food procurement).

This compilation feeds directly into the **Sodium Lives Saved Calculator** (next section) and determines how intervention effects are attributed to the correct sodium source.

Key method: Where country-specific data are unavailable, discretionary share is predicted from GDP per capita using the Bhat/Marklund regression (Bhat et al. 2020). Processed share is predicted from discretionary share. Floors: 10% discretionary (USA), 2% packaged (Ethiopia).

Sodium Lives Saved Calculator

Calculator Overview

Purpose: Estimate national-scale lives saved from a portfolio of sodium-reducing interventions.

Six intervention areas: (A) Public food procurement, (B) Behavior change / mass media, (C) Low-sodium salt substitutes, (D) Front-of-pack labeling, (E) Mandatory salt targets, (F) Fiscal policy / taxation.

Source disaggregation: Sodium intake split into discretionary, packaged, and outside-the-home (with $\sim \frac{1}{3}$ of outside-home = institutional). Each intervention acts on its target source.

Multiplicative combination: Effects within a source are combined multiplicatively, not additively, to prevent $\{>\}$ \$100% reductions. E.g., tax (13.7%) + FOPL (15%) on packaged $\Rightarrow 1 - (1 - 0.137)(1 - 0.15) = 26.6\%$.

Discretionary share: Predicted from GDP per capita (Bhat et al. 2020); floors at 10% (USA) and 2% packaged (Ethiopia).

Packaged-food growth: Projected using Vandevijvere et al. 2019 regional growth rates (Vandevijvere et al. 2019), capped at Singapore (4.3 g Na/day from packaged).

Demo: https://dcp-uw.shinyapps.io/salt_demo/ **Code:** <https://github.com/Disease-Control-Priorities/sodium>

Calculator Overview: Summary of Effect Sizes (Table 1)

Intervention	Source targeted	Study	Δ mg/day	Δ (%)	Time horizon
A. Public food procurement	Institutional	Gase 2011	−233	7%	Modeled
B. Behavior change / media	Discret. + Packaged	Do 2016	—	5%	1 year
C. LSSS	Discretionary	Huang 2026	—	TBD	TBD
D. Front-of-pack labeling	Packaged	Taillie 2024	−73.3	13.8%	3 years
E. Mandatory salt targets	Packaged	Strauss-Kruger 2022	−480	15.2%	4.56 years
F. Fiscal policy / taxation	Packaged (food categories)	Saxena 2025	−100	10%	20 years

Low-Sodium Salt Substitutes (LSSS)

Source targeted: Discretionary salt.

Study: Huang et al. (2026) — preprint (Matti et.al). Five scenarios:

1. Worldwide replacement of **all salt** (discretionary + non-discretionary)^{a, b}
2. Worldwide replacement of **discretionary salt only**^{a, b}
3. Worldwide replacement of non-discretionary salt only^{a, b}
4. Replacement of discretionary salt for people with **diagnosed** hypertension^{a, c}
5. Replacement of discretionary salt for people with **treated** hypertension^{a, c}

^a Modeled BP-lowering benefits. ^b Includes hyperkalemia harms in CKD G3–G5. ^c Assumes clinical screening prevents use in 20% of CKD G3–G5.

Working decision: Focus on scenarios **4, 5** (discretionary-only). Pending confirmation.

Sources of Sodium by Country

Country	Source	% Discret.	% Packaged	% Outside
South Africa	Charlton et al. 2005	33–46%	54–68%	—
Vietnam	Ha (PhD thesis, 2014)	13.5%	85.8%	—
Philippines	Amara et al. 2021	7.8–8.5%	90.4–93.3%	—
Argentina	Elorriaga et al. 2017	—	—	—
<i>Pending: Panama, Paraguay, Peru, Thailand</i>				

Sources: (Charlton, Steyn, et al. 2005; Ha 2014; Amaraño et al. 2021; Elorriaga et al. 2017). Where country-specific data are unavailable, discretionary share is predicted from GDP per capita using the Bhat/Marklund regression (Bhat et al. 2020); packaged share predicted from discretionary share.

Projected Growth of Packaged-Food Sales

Source: Vandevijvere et al. (2019) — change in ultra-processed food sales volume, 2002–2016.

Region	Change in UPF sales volume
South and Southeast Asia	+67.3%
North Africa and the Middle East	+57.6%
Central and East Asia	+10.0%
Western Europe	−3.8%
North America and Australasia	−2.2%

Used to project rising packaged-food share over the modeling horizon.

Cap: Singapore’s 4.3 g Na/day from packaged foods serves as the ceiling.

Implication: Countries in South/SE Asia will see a growing share of sodium from packaged sources, increasing the impact of reformulation, labeling, and taxation interventions over time.

SHAKE 2.0 / Revamped SHAKE Package

The WHO SHAKE package (2026 revision) now incorporates **low-sodium salt substitutes** alongside the original five pillars:

1. **Surveillance** — population sodium monitoring
 2. **Harness Industry** — food reformulation (mandatory/voluntary targets)
 3. **Adopt Standards** — front-of-pack labeling, marketing restrictions
 4. **Knowledge** — consumer information and education campaigns
 5. **Environments** — healthier food settings (institutional procurement)
- + **Low-Sodium Salt Substitutes (LSSS)** — newly added to the SHAKE package, reflecting the WHO guideline on LSSS and growing trial evidence

The largest and most sustained sodium reductions occur when SHAKE components are implemented as a **package**. Our modeling evaluates individual and combined policy bundles mapped to these components, now including LSSS as a sixth intervention lever.

Interventions Mapped to SHAKE Components

	Intervention	SHAKE Pillar	Source
A	Public food procurement	Environments (E)	Gase 2011
B	Behavior change / media	Knowledge (K)	Do 2016
C	LSSS	LSSS (+)	Huang 2026
D	Front-of-pack labeling	Adopt Standards (A)	Taillie 2024
E	Mandatory salt targets	Harness Industry (H)	Strauss-Kruger 2022
F	Fiscal policy / taxation	Harness Industry (H)	Saxena 2025

Interventions D, E, F target **packaged foods**; A targets **institutional** settings; B and C target **discretionary** sodium. **Surveillance??**

Team Discussion

Pending Decisions

1. **LSSS scenarios** — Confirm we model scenarios 4 and 5 (discretionary-only: all adults / diagnosed HTN / treated HTN).
2. **Mass media** — In or out? Laura: RTSL funds policy change, not media for behavior change. SHAKE framing argues for inclusion in the paper; reinvestment case argues for exclusion.
3. **Heterogeneity of observation periods** — Effect sizes measured over very different time windows (1 yr → 20 yrs). How do we harmonize? Options: (a) assume effects are end-state, apply at full scale-up; (b) apply linearly over observed window; (c) use a country-tier heuristic (low/middle/high).
4. **Source-of-sodium estimates** — How hard do we push for country-specific shares vs. fall back on Marklund's GDP regression with floors/ceilings? Flag condiment-heavy countries where the packaged-food transition is escalating.
5. **Integration with Matti's et.al** — Confirm we use his discretionary-vs-GDP regression and BP-effect inputs; clarify co-authorship/coordination so we do not scoop the *BMJ* paper.
6. **Bloomberg renewal deliverable** — What does the renewal application need from us, and by when, that is distinct from the paper?
7. **Timeline** — Original end-of-June paper submission is now +1–2 months for LSSS integration. New target?

Appendix

A. Public Food Procurement

Source targeted: Institutional share of outside-home consumption.

Selected study: Gase et al. (2011) — modeling/simulation study, LA County.

Policy simulated: Two approaches — (1) mandatory reformulation (reduce sodium content of all foods served); (2) choice architecture + incentives (labeling, promotion, subsidies, lower-Na alternatives).

Effect size: Weighted average across adult settings = **-233 mg/day ($\approx 7\%$)** of 3,329 mg baseline. Applies only to settings where individuals eat $\geq \$1$ meal/day, $\geq \$5$ days/week, $\geq \$8$ weeks.

Source-of-sodium adjustment: 233 mg = 20.6% reduction in sodium content of public meals. For a country where $x\%$ of sodium comes from institutional meals, population-level impact = $0.206 \times x\%$.

Caveats: Modeled, not observed; US-specific setting; very small affected population share.

B. Behavior Change / Mass Media

Source targeted: Discretionary + packaged.

Candidate study 1: Do et al. (2016) — COMBI Vietnam. Multi-component intervention (mass media + school + community + high-risk targeting). **5% decrease** in mean sodium excretion after 1-year intervention.

Candidate study 2: Collins et al. (2014) — UK economic evaluation. Advertising campaign promoting healthy eating. **2% [1–5%]** effect size used in their model. Provenance of these numbers could not be verified in the source references.

Open question

Laura's noted that RTSL funds **policy change, not mass media for behavior change**. Should this intervention be included in the model? SHAKE framing argues for inclusion; reinvestment case argues for exclusion.

C. Low-Sodium Salt Substitutes (LSSS)

Source targeted: Discretionary salt.

Study: Huang et al. (2026) — preprint (Marklund/TGI/Polly). Five scenarios:

1. Worldwide replacement of **all salt** (discretionary + non-discretionary)^{a, b}
2. Worldwide replacement of **discretionary salt only**^{a, b}
3. Worldwide replacement of non-discretionary salt only^{a, b}
4. Replacement of discretionary salt for people with **diagnosed** hypertension^{a, c}
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^a Modeled BP-lowering benefits. ^b Includes hyperkalemia harms in CKD G3–G5. ^c Assumes clinical screening prevents use in 20% of CKD G3–G5.

Working decision: Focus on scenarios **4, 5** (discretionary-only). Pending confirmation.

C. LSSS: Deaths Averted by Scenario

Country	Scenario 2 (Discret., all)	Scenario 4 (Discret., dx HTN)	Scenario 5 (Discret., tx HTN)
Global	1,860,000	600,000	470,000
China	689,246	205,936	169,409
India	277,196	54,096	46,445
Nigeria	13,214	3,257	2,047
Ethiopia	8,232	1,406	811
Bangladesh	36,044	7,510	5,931
Malaysia	8,798	3,200	2,930
Vietnam	36,831	10,672	7,530
Thailand	17,714	5,710	5,086
Philippines	32,899	9,106	6,533

Source: Huang et al. (2026). Scenarios 2, 4, 5 model discretionary-salt replacement only; scenarios 4 and 5 restrict to diagnosed/treated hypertensives respectively.

D. Front-of-Pack Labeling

Source targeted: Packaged (no adjustment needed — estimate already from packaged-food purchases).

Selected study: Taillie et al. (2024) — Chile FOPWL, observed.

Policy: Mandatory black octagonal warning labels for Na, sugar, saturated fat, and calories. Includes school sales bans and marketing restrictions on labeled products.

Population: 2,844 urban Chilean households, 138,391 household-months, 13 of 16 regions. July 2013 – June 2019.

Effect size: **−73.3 mg/capita/day (−13.8%)** in sodium purchases.

Caveat: Purchases-to-intake translation is uncertain. Nadia's US/Canada work used purchases directly.

Alternative: Mexico 2026 study found −7% (95% CI −13%, −2%) in packaged-food/beverage *sales* $\approx$ −38 mg/capita/day.

E. Mandatory Salt Targets

Source targeted: Packaged foods.

Policy: South African mandatory national sodium reduction legislation (2013); maximum sodium limits across 13 categories of processed foods; targets implemented 2016 and 2019.

Selected study: Strauss-Kruger et al. (2022) — African-PREDICT. $N = 311$ healthy 20–30-year-olds, 24-hr urine. Baseline 2013–2017 vs. follow-up 2018–2020; mean follow-up 4.56 yrs.

Effect size: **–15.2% sodium/day** (derived). Baseline/follow-up straddle target dates → likely **underestimates** true effect.

Source adjustment: 62% of SA sodium from packaged food (Charlton, Steyn, et al. 2005). Raw adjustment yields 24.5% reduction relative to packaged source; conservative estimate = 20%.

Alternative SA studies: Gaziano 2025 (–8.4%, rural, ≥ 40 yrs, spot urine) (Gaziano et al. 2025); Charlton 2021 (–18.4%, national, methodological concerns) (Charlton, Ware, et al. 2021).

UK targets: He 2013: –15% over 7 yrs, voluntary (He, Pombo-Rodrigues, and MacGregor 2013); Song 2023: –19.2% over 11 yrs (Song et al. 2023).

Mexico modeling: Muciño-Sandoval 2025: –25.2% phase 1, –28.6% phase 2 (Muciño-Sandoval, Barrientos-Gutiérrez, et al. 2025) — justifies not using Gaziano’s 8%.

F. Fiscal Policy / Taxation

Source targeted: Packaged (no adjustment needed).

Selected study: Saxena et al. (2025) — Philippines, modeled. Elasticity for food groups.

Policy: 20% tax on foods high in salt and sugar; WHO WPRO nutrient-profile model; 100% pass-through; industry reformulation ($\leq 50\%$ above threshold).

Population: Adults of the Philippines, aged ≥ 25 years.

Effect size: **−100 mg Na/day (−10%)** at year 20.

Caveat: High relative to other estimates; tax does not include fat.

Alternative studies:

- India (Roche et al. 2026): −1.63% over 30 yrs (40% GST, non-standard tax structure).
- Australia (Davies et al. 2026): −110 mg/day ($\approx -3.22\%$) over lifetime of current population.
- Australia (Cobiac et al. 2017): −67 mg/day under \$0.30/g Na excise.
- Dodd et al. (2020) systematic review: modeling studies predict 1–11% decrease in salt intake.

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